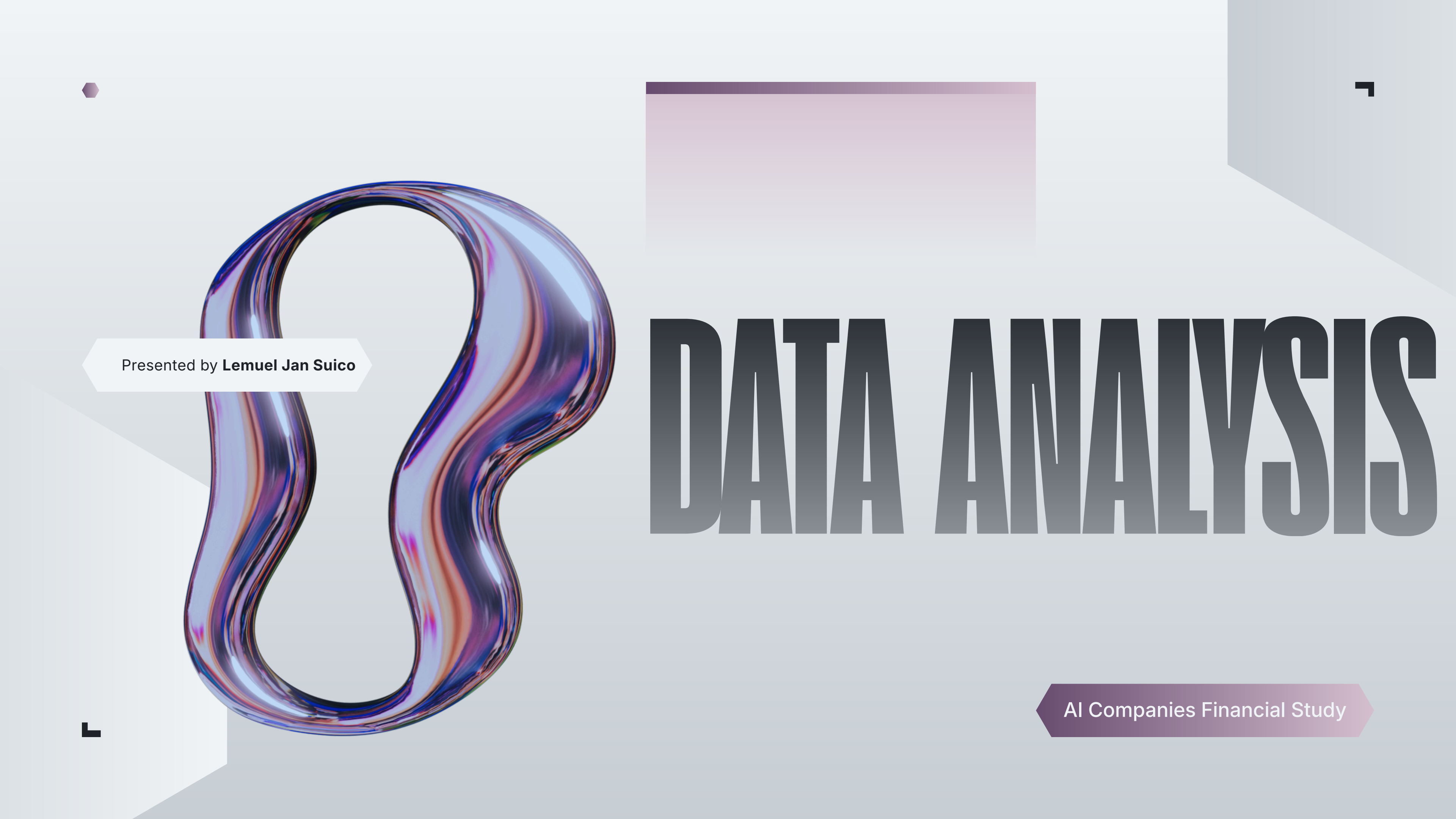




Presented by **Lemuel Jan Suico**

DATA ANALYSIS

AI Companies Financial Study

PROJECT OVERVIEW

AI Industry Data Analysis

This project focuses on analyzing AI industry data stored in the `aidata.csv` dataset using Python-based data analysis tools. The study was conducted using Pandas for data manipulation, Matplotlib for visualization, and Seaborn for advanced statistical plotting.

The Goal

The goal of the analysis is to understand how AI-related revenue, research and development spending, company events, and stock market impact are connected within the technology industry.

Sourced

The dataset, sourced from [Kaggle](#), contains 10,959 records on company performance, AI revenue, growth, R&D investment, and market activity used to analyze how AI investment contributes to business growth.

```
analysis.py X  
analysis.py > ...  
  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns  
import os  
  
os.makedirs("output", exist_ok=True)  
  
df = pd.read_csv("aidata.csv")  
  
print("FIRST 5 ROWS")  
print(df.head())
```

The dataset contains several important variables that represent financial and operational indicators within AI-focused companies.

Datasets Description

The Date column records when each entry occurred, while the Company column identifies the organization associated with the data. The R&D_Spending_USD_Mn column represents the amount spent on research and development activities, and the AI_Revenue_USD_Mn column measures revenue generated from AI-related operations.

The dataset also includes AI_Revenue_Growth_%, which tracks revenue growth percentages over time, along with the Event column that records major announcements or business activities. Lastly, the Stock_Impact_% column reflects how company events and AI performance affected stock market behavior.

DATASET OVERVIEW

Datasets in CSV format

```
aidata.csv > data
1 Date,Company,R&D_Spending_USD_Mn,AI_Revenue_USD_Mn,AI_Revenue_Growth_%,Event,Stock_Impact_%
2 2015-01-01,OpenAI,5.92,0.63,-36.82,, -0.36
3 2015-01-02,OpenAI,5.41,1.81,80.59,, 0.41
4 2015-01-03,OpenAI,4.5,0.61,-38.88,, 0.23
5 2015-01-04,OpenAI,5.45,0.95,-5.34,, 0.93
6 2015-01-05,OpenAI,3.4,1.48,48.45,, -0.09
7 2015-01-06,OpenAI,5.47,1.08,8.32,, 0.74
8 2015-01-07,OpenAI,6.95,0.67,-32.95,, -0.8
9 2015-01-08,OpenAI,5.06,1.43,43.49,, -0.09
10 2015-01-09,OpenAI,5.26,0.33,-67.28,, -0.72
11 2015-01-10,OpenAI,2.0,0.74,-25.8,, 0.5
12 2015-01-11,OpenAI,5.34,1.88,88.19,, -0.4
13 2015-01-12,OpenAI,6.1,0.33,-66.82,, -0.76
14 2015-01-13,OpenAI,3.09,0.74,-25.84,, -0.35
15 2015-01-14,OpenAI,5.18,0.72,-27.93,, 0.31
16 2015-01-15,OpenAI,6.16,0.95,-4.66,, 0.53
17 2015-01-16,OpenAI,4.68,0.88,-11.77,, -0.82
18 2015-01-17,OpenAI,4.19,0.61,-38.76,, -0.21
19 2015-01-18,OpenAI,3.46,1.16,15.85,, -0.41
20 2015-01-19,OpenAI,4.12,1.76,75.87,, 0.56
21 2015-01-20,OpenAI,4.37,0.91,-9.27,Cloud AI Launch,2.43
22 2015-01-21,OpenAI,5.58,1.09,9.27,, 0.36
23 2015-01-22,OpenAI,4.87,1.21,21.4,, -0.37
24 2015-01-23,OpenAI,5.46,0.94,-5.58,, -0.78
25 2015-01-24,OpenAI,5.75,0.35,-64.64,, -0.88
26 2015-01-25,OpenAI,4.3,0.77,-22.72,, -0.68
27 2015-01-26,OpenAI,6.7,0.41,-58.59,, -0.52
28 2015-01-27,OpenAI,5.85,0.83,-16.94,, -0.87
29 2015-01-28,OpenAI,4.13,1.15,15.13,, -0.56
```



Before performing the analysis, the dataset underwent a cleaning process to improve data quality and consistency.

DATASETS CLEANING

Capital Flows into AI

Data Cleaning

The Date column was converted into datetime format to ensure proper handling of dates. Missing values in the Event column were replaced with the label No Event to maintain consistency throughout the dataset.

```
print("\nSTATISTICS")
print(df.describe())

df["Date"] = pd.to_datetime(df["Date"])

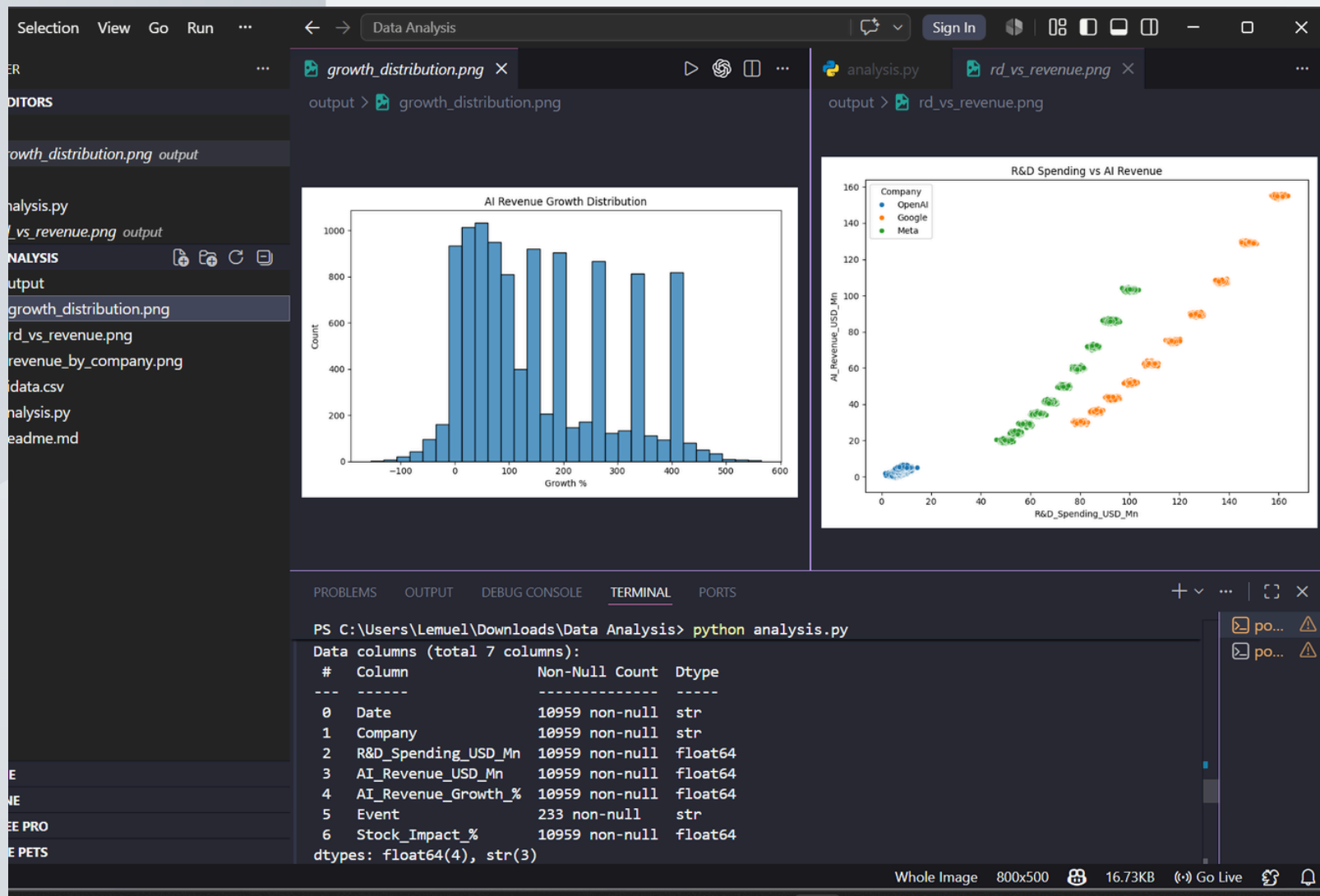
df["Event"] = df["Event"].fillna("No Event")

df = df.drop_duplicates()

print("\nCLEANED DATA SHAPE:", df.shape)
```

Duplicate rows were removed to prevent repeated records from affecting the analysis.

After cleaning, the dataset became more reliable for statistical exploration and visualization.



The analysis of R&D spending showed that companies investing more in research and innovation generally achieved stronger AI revenue performance. Google also had the highest R&D spending, suggesting a strong relationship between investment and profitability.

Revenue growth and stock market analysis showed that AI business performance can fluctuate over time due to market conditions and company events. The results also indicated that AI-related developments usually create moderate positive reactions from investors.

The exploratory data analysis revealed patterns in AI revenue and company investment behavior. Google recorded the highest average AI revenue, followed by Meta and OpenAI, showing that larger technology companies currently dominate AI-driven revenue generation.

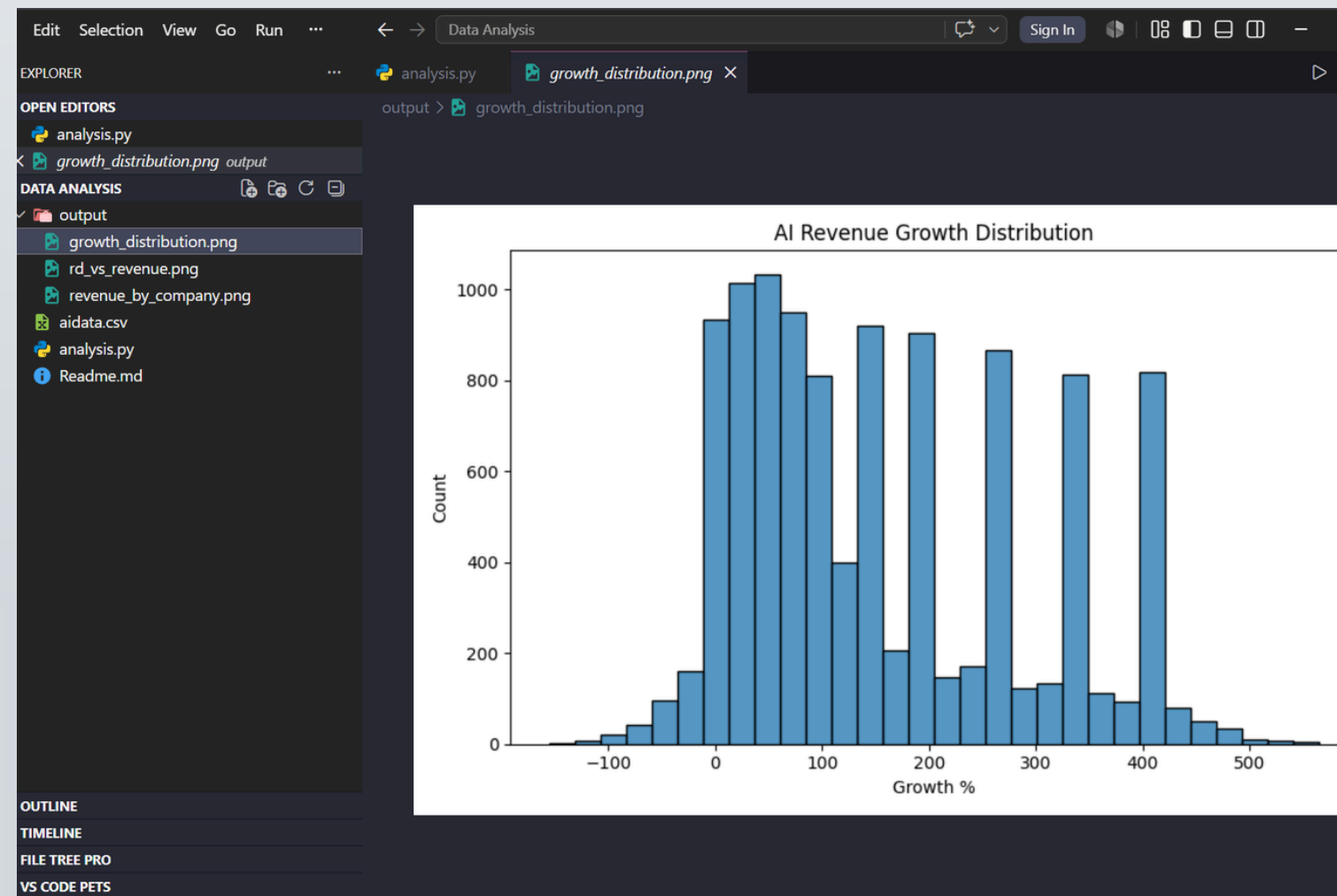
EXPLORATORY DATA ANALYSIS

AI Industry Data Analysis

VISUAL GENERATION

Visualizations

These visualizations made the analysis more accessible and helped transform numerical data into meaningful business insights.



Several visualizations were created during the project to improve interpretation of the data. A bar chart was used to compare average AI revenue among companies, allowing quick identification of the strongest performers.

Visuals

A scatter plot was generated to visualize the relationship between R&D spending and AI revenue, helping demonstrate the positive correlation between investment and profitability. A histogram was also created to show the distribution of AI revenue growth percentages, making it easier to observe the variability and trends within the dataset.

Companies with higher R&D spending consistently generated stronger AI revenue, showing that innovation investment greatly contributes to financial success. Google led in both AI revenue and R&D spending, highlighting the link between research investment and market performance.

Observations

The analysis revealed that AI revenue growth can fluctuate depending on market conditions and company activities. AI-related announcements and developments also influence stock market behavior, though the overall effect is moderate.

```
avg_revenue = df.groupby("Company")[\n\nprint("\nAVERAGE AI REVENUE")\nprint(avg_revenue)\n\navg_rd = df.groupby("Company")["R&D_\n\nprint("\nAVERAGE R&D SPENDING")\nprint(avg_rd)\n\nplt.figure(figsize=(10, 5))\n\navg_revenue.sort_values().plot(kind=\n\nplt.title("Average AI Revenue by Com\nplt.xlabel("Company")\nplt.ylabel("Revenue (USD Mn)")\nplt.tight_layout()\n\nplt.savefig("output/revenue_by_compa\nplt.close()\n\nplt.figure(figsize=(8, 6))
```

KEY INSIGHTS

Findings

Overall, artificial intelligence remains a major growth driver in the technology industry. Companies that invest heavily in research and development are more likely to achieve stronger performance and competitive advantage.

INTERPRETATIONS

Recommendations

Based on the results of the analysis, companies should continue investing in AI research and development to maintain competitive advantage and support long-term revenue growth. Monitoring fluctuations in AI revenue growth may also help organizations identify market opportunities and potential risks more effectively.

Project Recommendation

Additional statistical methods such as correlation heatmaps and regression analysis could also provide deeper insights into the relationship between AI investment and business performance.

Practical Implications

Future improvements to this project could include implementing machine learning models for revenue prediction, developing interactive dashboards using Streamlit, and applying time-series forecasting techniques to analyze long-term trends.

OUT DEBUG CONSOLE <u>TERMINAL</u> PORTS			
C:\Users\user\Downloads\Data Analysis> python analysis.py			
Investing_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth_%	St
10959.000000	10959.000000	10959.000000	
65.184504	44.126571	159.395988	
47.918247	41.639356	135.462185	
1.570000	-0.550000	-155.430000	
8.640000	3.610000	43.870000	
70.960000	35.220000	133.750000	
99.600000	71.680000	258.300000	
163.830000	155.960000	565.500000	



This project demonstrates how Python data analysis tools can be used to explore AI industry performance effectively.

Using Pandas, Matplotlib, and Seaborn allowed:

- efficient data cleaning
- statistical analysis
- professional visualization
- business insight generation

REALIZATIONS

Conclusion

The analysis revealed a strong relationship between R&D spending and AI revenue performance, while also highlighting the growing financial impact of AI innovation across major technology companies.

THANK YOU

Closing

Lemuel Jan Suico

This project is available in my GitHub repository. You can access the repository through the GitHub profile link provided on the website's social links section by clicking the GitHub icon button.

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